

General Relativity

Pre-Final Examination Spring 2021 Total Possible Score: 100

Name: _____ Class: _____ Student number: _____

1. (15 points) From the geodesic equation, Eqn. (3.44) in Carroll,

show that the quantity

$$X \equiv g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}$$

maintains a constant value along a geodesic. State clearly any assumptions you make.

2. (30 points)

- (a) Show that Maxwell's equations, $\partial_\mu F^{\nu\mu} = J^\nu$, lead to current conservation in Minkowski spacetime.
(b) If you follow the exact same procedure to show that current is conserved in curved spacetime, you will quickly run into a problem. Can you explain why?
(c) I assume you have written down a valid equation in part (b).
Now split the left-hand side of this equation up into two terms:
the field strength tensor times something which is totally symmetric,
and the field strength tensor times something which is totally anti-symmetric.
(d) Now write an expression involving the Riemann curvature tensor (assume torsion free).
(e) Next introduce the Ricci tensor where appropriate.
(f) Now show that current is conserved in curved spacetime.
(g) (a bit tricky) Actually, there is an additional complication if we write the field strength tensor in terms of the 4-vector potential. Then there is an apparent ambiguity (歧义) in going from the technique in part (a) to the technique in part (b). Can you say what this is?
(The resolution of this ambiguity is very similar to the procedure you have just gone through. Do not attempt to do this!)

3. (20 points) Consider the line element given by

$$ds^2 = e^{-2\xi} dt^2 - e^{+2\xi} (dx^2 + dy^2 + dz^2) .$$

Find the Christoffel symbols and the geodesic equations.

4. (35 points) Consider the line element given by

$$ds^2 = -(dudw + dwdu) + \alpha^2 dx^2 + \beta^2 dy^2 ,$$

where α and β are both functions of (only) u .

- (a) Find all Christoffel symbols.
(b) Find the non-zero terms of the Riemann curvature tensor.

Hints: recall from lecture that I said to look for terms of the form "a-b-a-b";
in addition here you will need to look at terms of the form "a-b-a-c".
Once you have found a term of the form "a-b-c-d" is either non-zero or zero, you are NOT required to investigate any term which is a permutation of the same set of indices, such as "b-a-c-d".

- (c) Find all terms of the Ricci tensor.
(d) Find the Ricci scalar.
(e) Write down a differential equation for α and β valid in a vacuum.
(f) IF this were Minkowski spacetime, find choices for α and β together with a suitable explicit transformation from canonical (Cartesian) coordinates to those given in the line element above. Confirm that the answers in part (b) change as expected.